



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
13-09-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

PAPER-2

Time : 3 Hours

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

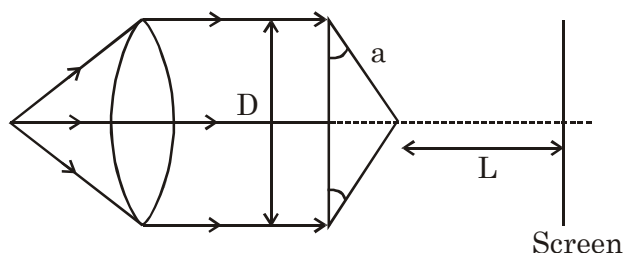
SECTION-I(i) : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
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Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. A shortwave transmitter operates at a frequency $\nu = 30$ MHz. A receiver is located at a distance L from it. Radiowaves reach the receiver after being reflected by ionospheric layers at altitudes of $h_1 = 100$ km and $h_2 = 300$ km. Find the location of maxima from the transmitter as the receiver is moved along the line connecting it to the transmitter. Assume earth to be flat:-
- (A) $300\sqrt{5}$ km (B) 0 km (C) $500\sqrt{3}$ km (D) None of these
-

Space for Rough Work

2. In a two slit interference pattern, light of wavelength 500 nm is used. The intensity at central maxima is I_0 . The intensity at a point P which is closer to one slit by 400 nm than other slit is I_1 . The intensity at a point Q where phase difference between the two waves is $\frac{2\pi}{3}$ rad is I_2 .
- (A) $I_0 > I_1 > I_2$ (B) $I_0 > I_2 > I_1$ (C) $I_1 \approx 0.64 I_0$ (D) $I_2 = 0.25 I_0$
3. A point source of light S is located at the focal point of a lens. A symmetric biprism with interior angle $a = 0.01$ rad and width $D = 6$ cm is situated behind the lens (figure). At what distance L from the prism can the largest number of interference bands be observed? How many bands N can be seen on this screen? What is the width W of the bands? The refractive index of the prism glass is $n = 1.5$, and the wavelength of the light is $\lambda = 0.5 \mu\text{m}$.



- (A) $L = 3$ m (B) $N = 60,000$ (C) $W = 0.1$ mm (D) None of these

Space for Rough Work

4. It is a common knowledge that we can't see clearly underwater with naked eyes. But if one wears goggles with air between eyes and goggle, we can see clearly. If instead of goggles, one used glasses with air between glasses and eye, (equiconvex or equiconcave) ($\mu = 1.5$) underwater :
- (A) A normal eye person will see clearly using equiconcave lens.
 (B) A normal eye person will see clearly using equiconvex lens.
 (C) A myopic (near sighted person) will see clearly using equiconvex lens.
 (D) A myopic person will see clearly using equiconcave lens.
5. A string of length L is clamped at both the ends. It is oscillating in 1st overtone. At $t = 0$, the string is straight. $x = 0$ is the left end of the string. Choose the combinations of x & t where speed is half the maximum speed at the antinode. Time period = T
- (A) $x = \frac{L}{4}$, $t = \frac{T}{3}$ (B) $x = \frac{L}{4}$, $t = \frac{T}{6}$ (C) $x = \frac{L}{3}$, $t = \frac{T}{4}$ (D) $x = \frac{L}{12}$, $t = \frac{T}{2}$
6. Two strings of same material are joined to form a large string and is stretched between rigid supports. The diameter of the second string is twice that of the first. It was observed in an experiment that the whole string was oscillating in 4 loops with a node at the joint. Find the possible lengths of the second string if the length of first string is 90 cm.
- (A) 15 cm (B) 90 cm (C) 135 cm (D) 45 cm

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

The concept of Doppler shift can also be applied to light waves. But according to the theory of relativity. It is immaterial whether source moves or the observer moves. If the observer moves with a velocity of v relative to the source towards the source, the frequency that he perceives is

$$f = \sqrt{\frac{c+v}{c-v}} f_0$$

But if the observer moves with a velocity of v relative to the source away from the source, the

frequency that he perceives is $f = \sqrt{\frac{c-v}{c+v}} f_0$

This is frequently used by police officers to find the velocity of a moving car. They point a radar gun towards a car moving towards or away from them.

7. If the wave reflected by moving car has shift in frequency Δf and f_0 is the original frequency, and $\frac{\Delta f}{f_0} = 3 \times 10^{-7}$ what is the speed of the car (approx.) ?

(A) 30 m/s (B) 45 m/s (C) 90 m/s (D) 22.5 m/s

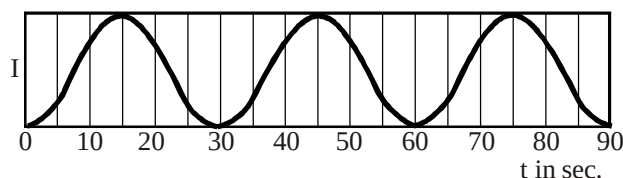
8. According to theory of relativity, the speed of light is same for all observers. If the source emits light of frequency 10^{15} Hz. An observer moving at a speed of $0.6c$ towards the source perceives a wavelength of

(A) 300 nm (B) 150 nm (C) 225 nm (D) 450 nm

Space for Rough Work

Paragraph for Questions 9 and 10

In the front of the upper slit of YDSE apparatus, a thin film of a liquid of refractive index 1.40 is placed. It is a hot day and the liquid starts evaporating from the surface. A beam of light at wavelength 560 nm is incident onto the YDSE apparatus and the intensity I at the centre of the screen is monitored. Figure gives intensity I as a function of time t . The intensity changes because of evaporation from the two sides of the film. Assume that the film is flat and has parallel sides. Also assume that the film's thickness decreases at a constant rate.

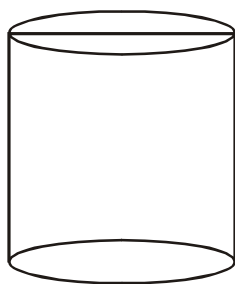


9. What can be the initial thickness of the film ?
 (A) $7 \mu\text{m}$ (B) $4.9 \mu\text{m}$ (C) $7.5 \mu\text{m}$ (D) $8.5 \mu\text{m}$
10. The rate of change of thickness can be
 (A) $\frac{140}{3} \text{ nm/sec}$ (B) 70 nm/sec (C) $\frac{28}{3} \text{ nm/sec}$ (D) $\frac{56}{3} \text{ nm/sec}$

Space for Rough Work

Paragraph for Questions 11 and 12

You have designed a new musical instrument of very simple construction. Your design consists of a metal tube with length L and diameter $L/10$. You have stretched a string of mass per unit length μ across the open end of the tube. The other end of the tube is closed. It produces the musical effect you're looking for, you want the frequency of the third-harmonic standing wave on the string to be the same as the fundamental frequency for sound waves in the air column in the tube. The speed of sound waves in this air column is v_s .



11. What must be the tension of the string to produce the desired effect ?
- (A) $\frac{\mu v_s^2}{4900}$ (B) $\frac{9\mu v_s^2}{400}$ (C) $\frac{\mu v_s^2}{3600}$ (D) None of these
12. For the tension calculated in Q. No. 11, what other harmonics of the string, if any, are in resonance with standing waves in the air column ?
- (A) 2,4,6, (B) 9,15, 21 (C) 4,8,12,..... (D) 5,10,15,.....

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

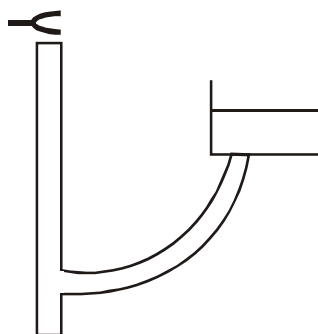
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●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A distant source of sound has frequency 850Hz. An observer is facing 90° away from the direction of the source. Estimate the phase difference (in radians) between the oscillations of her left and right eardrums. Speed of sound in air = 340 ms^{-1} . Assume the distance between both the ears is 25 cm.
2. An antenna emits a plane electromagnetic wave with frequency $\nu = 3 \times 10^{11} \text{ Hz}$ in the direction perpendicular to a reflecting screen. An electric field intensity meter establishes the result of interference of the incident and reflected waves as they move between the antenna and the screen. Determine the minimum thickness (in mm) of a dielectric plate attached flat against the screen such that the meter shows a minimum electric field intensity if, before the plate is attached, it was located at a spot with maximum intensity. The refractive index of the plate is $n = 1.5$; ignore absorption by the plate and reflection from it.

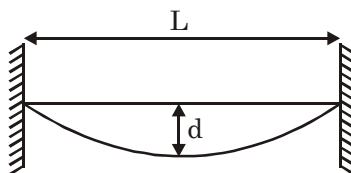
Space for Rough Work

3. Travelers in the desert sometimes observe what appears to be a sea or lake. At what distance (in m) from the observer does such a mirage appear? Assume that the speed of light near the ground in the desert varies according to the formula $c(z) = c_0(1 - az)$, where c_0 is the speed of light at the ground and z is the altitude above the ground. Assume rays were travelling parallel to ground near the ground. [$h = 2\text{m}$, $a = \frac{1}{2500} \text{m}^{-1}$, (where h is height of man eye from ground)]
4. In resonance column experiment a tuning fork of frequency $f = 400 \text{ Hz}$ is held above the pipe as shown in figure. The reservoir is raised and lowered to change the level of water and thus the length of the column of air in the tube. The area of cross section of the reservoir is 6 times that of the pipe. Initially, the reservoir is kept so that the pipe is full up to the brim. Tuning fork is sounded and the reservoir is lowered. When the reservoir is lowered by 21 cm, first resonance is recorded. When the reservoir is lowered further by 49 cm the second resonance is heard. Find the speed of sound in air (in m/s).



Space for Rough Work

5. A wire having mass per unit length μ and length L is fixed between two fixed vertical walls at a separation L . Due to its own weight the wire sags. The sag in the middle is d ($\ll L$). Assume that tension is practically constant along the wire, owing to its small mass. Calculate the speed (in m/s) of the transverse wave on the wire. (Take : $L = 1\text{m}$, $d = 2\text{mm}$)



6. Two identical violin strings, when in tune and stretched with the same tension, have a fundamental frequency of 440.0 Hz . One of the strings is retuned by increasing its tension. When this is done, 1.5 beats per second are heard when both strings are plucked simultaneously at their centers. By what percentage was the string tension changed ?

Space for Rough Work

PART-2 : CHEMISTRY**SECTION-I(i) : (Maximum Marks: 24)**

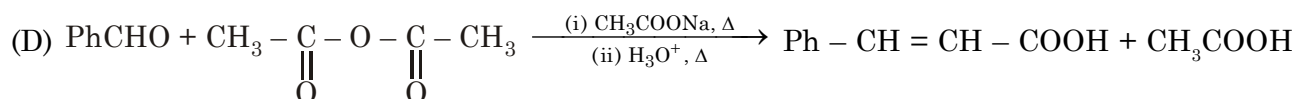
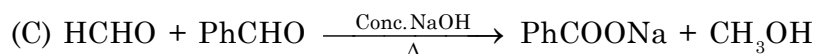
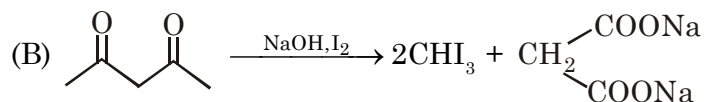
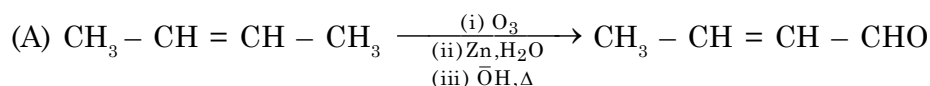
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-
1. Which of following order is correct.
(A) K_2O_3 is a mixed oxide
(B) K_2O is used in submarine to absorb CO_2
(C) KOH is preferred to absorb CO_2 compared to NaOH
(D) $|\Delta H_f|$ order for alkali metals fluoride increases down the group
 2. Which of the following characteristic is **CORRECT** when alkali metal is dissolved in liquid ammonia.
(A) Deep blue solution is observed
(B) Obtained solution is paramagnetic in nature
(C) Obtained solution conduct electricity
(D) Obtained solution acts as reducing agent
-

Space for Rough Work

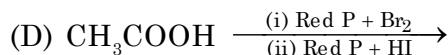
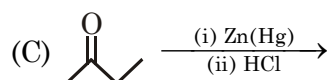
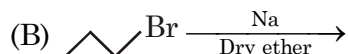
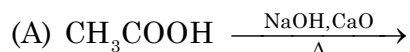
3. Select the correct statement (s)-
- (A) On increasing mass number, binding energy first increases then decreases
- (B) On increasing mass number, binding energy per nucleon first increases then decreases.
- (C) Specific activity of ${}_{92}^{238}\text{U}_3\text{O}_8$ is more than ${}_{92}^{238}\text{UF}_6$
- (D) Isotopes of same element may be involved in different disintegration series.
4. An ideal solution is formed by mixing one mole each of benzene and toluene. At 300K, vapour pressure of pure benzene and toluene are 100 mm Hg and 40 mm Hg respectively. If pressure over mixture is reduced at same temperature, then correct(s) option is/are -
- (A) Pressure at which first trace of vapour formed is 70 mm Hg
- (B) Mole fraction of benzene in of first trace of vapour formed is 0.714
- (C) Mole fraction of benzene in of last trace of liquid drop left is 2/7
- (D) Pressure at which half of total moles of liquid will vaporize is 63.25 mm Hg

Space for Rough Work

5. Major product of which of the following reactions are correctly matched :



6. Which of the following reaction will give an alkane having different number of carbon atoms as compare to the reactant :



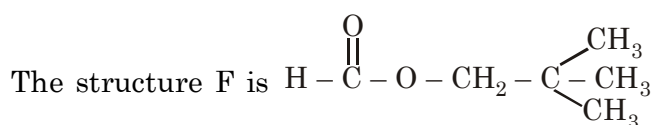
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

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Paragraph for Questions 7 and 8

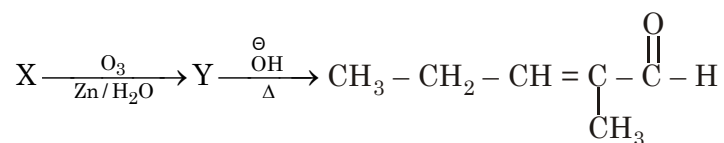
An alkene (A) C_6H_{12} gives two different aldehyde B and C on reductive ozonolysis which further gives product D and E by cross cannizaro reaction in presence of conc. NaOH. Finally product D and E gives an ester (F) in presence of conc. H_2SO_4 .



7. The structure of A is?

- (A) $CH_3 - CH_2 - CH = CH - CH_2 - CH_3$ (B) $\text{>CH} = CH_2$
 (C) $\text{>CH} = CH - CH_2 - CH_3$ (D) $\text{>CH} = CH - CH_3$

8. Compound X is isomer of compound A.



Compound X and compound A are?

- (A) Position isomers (B) Functional isomers
 (C) Chain isomers (D) Geometrical isomers

Space for Rough Work

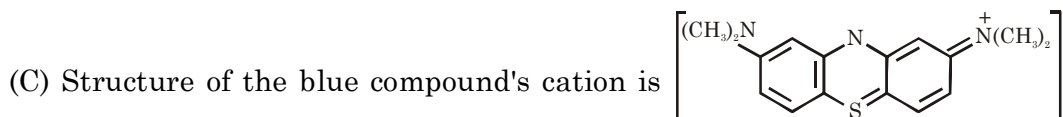
Paragraph for Questions 9 and 10

Qualitative analysis of inorganic salt is carried out through the reactions which are easily perceptible to our senses such as sight and smell. Such reactions involve:

- (a) Formation of a precipitate
- (b) Change in colour
- (c) Evolution of gas etc.

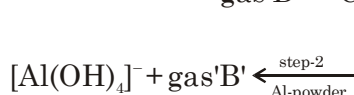
9. N, N-dimethyl p-phenylenediamine + Na_2S + 'R' $\xrightarrow{\text{H}^+}$ deep blue colouration which of the following statement is incorrect?

- (A) R is FeCl_3
- (B) On the place of Na_2S , H_2S also gives same observation



- (D) None of these

10. Compound 'A' + conc. NaOH (excess) $\xrightarrow{\text{step-1}}$ gas 'B' + 'C' solution



Some compounds are given here.

- (I) NH_4Cl (II) NH_4NO_3 (III) $(\text{NH}_4)_2\text{SO}_4$ (IV) NH_4NO_2

Choose the correct statement from the following options.

- (A) gas B in step-1 can be obtained from II, IV only
- (B) gas B in step-2 can be obtained from II only
- (C) gas B in both steps can be obtained from II and IV only
- (D) gas B turns blue litmus into red

Space for Rough Work

Paragraph for Questions 11 and 12

Conformation is the momentary arrangement of atoms in three dimension surrounding the sp^3 hybridised carbon atom. For C_2H_6 mainly two conformers are possible viz. eclipsed & staggered. Staggered is more stable conformer.

The standard free energy difference for two conformers is 2.76 kcal/mol at 300K

ΔG_f^0 (staggered C_2H_6) = -7 kcal/mol at 300 K

ΔH_f^0 (natural C_2H_6) = -27.9724 kcal/mol at 300 K (Take $\log_e x = 2.3 \log_{10} x$)

11. What % of staggered C_2H_6 is present in natural C_2H_6 at 300 K ?
(A) 1% (B) 99% (C) 10% (D) 90%
12. What is ΔS_f^0 of natural C_2H_6 at 300 K
(A) 35 cal/k-mol (B) -35 cal/mol-k
(C) -70 cal/k-mol (D) -70 k-cal/k-mol

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

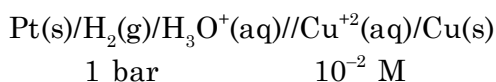
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. Consider the cell



Measured E.M.F. of given cell is 1 V. Calculate pOH of anodic half solution at 25°C

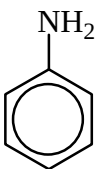
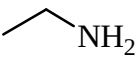
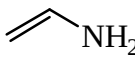
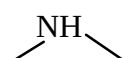
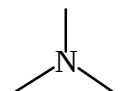
$$\left[E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = 0.34\text{V, take } \frac{2.303 RT}{F} = 0.06 \right]$$

2. 100 ml of 0.1 M H_3PO_4 is titrated with 0.5 M NaOH solution till 2nd equivalence point. Then resultant solution was mixed with 10 ml of 0.5 M HCl solution. What will the pH of final solution after addition of HCl ?

$$K_1 = 10^{-3} \text{ M} ; K_2 = 10^{-8} \text{ M} ; K_3 = 10^{-13} \text{ M}$$

Space for Rough Work

3. How many reagents or solution which can be use to differentiate propanal and butanone.
- | | |
|----------------------------------|-------------------------|
| (i) Ammonical silver nitrate | (ii) Fehling solution |
| (iii) $\text{NaOH} + \text{I}_2$ | (iv) Sodium bi sulphite |
| (v) 2,4-dinitro phenyl hydrazine | (vi) Lucas reagent |
| (vii) Benedict solution | (viii) Schiff's reagent |
| (ix) Hinsberg reagent | |

4. (i)  (ii) $\text{CH}_3\text{-NH}_2$ (iii)  (iv) 
- (v)  (vi)  (vii) $\text{Ph-CH}_2\text{-NH}_2$

In the above compound how many can synthesis by Gabriel method.

5. Find the total number of optically active isomer for $[\text{Co}(\text{NH}_3)(\text{gly})_2\text{SCN}]$.
6. $\text{Na}_2\text{CO}_{3(\text{aq. solution})} + \text{reagent 'A'} \longrightarrow \text{white ppt.}$
 Find the number of 'A' from the following reagent
 CuSO_4 , $(\text{BaCl}_2 + \text{CH}_3\text{COOH})$, $\text{Pb}(\text{OAc})_2$, ZnSO_4 , $\text{Hg}(\text{NO}_3)_2$, $\text{Hg}_2(\text{NO}_3)_2$, CaCl_2 , $(\text{SrCl}_2 + \text{HCN})$

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks: 24)**

- This section contains **SIX** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
1. A tangent is drawn at any point $P(t)$ on the parabola $y^2 = 8x$ and a point $Q(\alpha, \beta)$ is taken on it from which pair of tangents QA and QB are drawn to the circle $x^2 + y^2 = 4$, where A & B are points of contact. Now if point $P(t)$ varies on the parabola then the locus of all such point of concurrency of the chord of contact AB is denoted by $S = 0$, then-
- (A) Equation of $S = 0$ is $2x + y^2 = 0$
(B) Equation of $S = 0$ is $x + y^2 = 0$
(C) Area bounded by $S = 0$ and the line $2x + 1 = 0$ is $\frac{2}{3}$ sq. units
(D) Tangents drawn at the extremities of intersection of $S = 0$ and the line $2x - y + 1 = 0$ to the curve $S = 0$ includes an angle 75° .
-

Space for Rough Work

2. A circle of radius 'r' passes through both foci and exactly four points on the ellipse with equation $x^2 + 16y^2 = 16$. The set of all possible values of r is an interval [a,b), then a + b is equal to -

(A) $5\sqrt{2} + 4$ (B) $\sqrt{17} + 7$ (C) $6\sqrt{2} + 3$ (D) $\sqrt{15} + 8$

3. The value of the integral $\int \frac{x(2-x)}{e^x(1+x^2e^{-x})^2} dx$ is -

(A) $\frac{e^{-x}}{e^{-x} + x^2} + C$ (B) $-\frac{e^x}{e^x + x^2} + C$ (C) $-\frac{e^{-x}}{e^{-x} + x^2} + C$ (D) $\frac{e^x}{e^x + x^2} + C$

(where C is constant of integration)

4. Which of the following statements are **correct**?

(A) The solution set of $[\sin^{-1}x] > [\cos^{-1}x]$ contains exactly one integer.

(B) Range of $[\sin x] + [\cos x]$ contains exactly two integers.

(C) Range of $\left[x + \frac{1}{2}\right] + \left[x - \frac{1}{2}\right] + 2[-x]$ contains exactly two integers.

(D) Range of $\left[\frac{1}{[x-3]}\right]$ contains exactly three integers

where $[.]$ = greatest integer function.

Space for Rough Work

5. If $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{d}| = 1$ such that $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = 1$ and $\vec{a} \cdot \vec{c} = \frac{1}{2}$, then which of the following can be correct ?
- (A) $\vec{a} \cdot \vec{b} = 0$ (B) $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$ (C) $\vec{a} \cdot (\vec{b} \times \vec{c}) \neq 0$ (D) $|\vec{a} + \vec{b} + \vec{c}| = \sqrt{4 + \sqrt{3}}$
6. Let $P_1 \equiv x + y + z = 6$ & $P_2 \equiv x + y + z = 11$ be the two parallel planes. Let three points A, B & C are such that their x, y & z co-ordinates can take only positive integral values, then -
- (A) if point A & origin lies on same side of plane P_1 , then A can take 10 different locations in space.
- (B) if point B & origin lies on same side of plane P_2 , then B can take 10 different locations in space.
- (C) if point B & origin lies on same side of plane P_2 , then B can take 120 different locations in space.
- (D) if point C lies between planes P_1 & P_2 , then it can take 100 different locations in space.

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 18)

- This section contains **THREE** paragraphs.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

Paragraph for Questions 7 and 8

Line $x + y = 6$ is tangent to a parabola 'S' at P(5,1) and M(3,-1) is foot of perpendicular from P on its directrix.

7. If coordinates of vertex of parabola 'S' are (α, β) , then $\alpha - \beta$ is equal to
(A) 1 (B) 2 (C) 3 (D) 4
8. Tangents are drawn to the parabola 'S' from M which touches the parabola at the points whose coordinates are -
(A) (3,7), (12,-2) (B) (11,-1), (8,2) (C) (3,7), (7,3) (D) (3,7), (11,-1)

Space for Rough Work

Paragraph for Questions 9 and 10

Consider an ellipse $E: \frac{x^2}{4} + \frac{y^2}{3} = 1$, S is its focus and line L is its corresponding directrix.

Curve 'C' is the locus of a variable point which moves on xy plane such that its distance from S and line L are equal.

9. Number of common tangent(s) of curve C and E is-
(A) 0 (B) 1 (C) 2 (D) 3
10. Number of normals of curve C which are tangents for ellipse E is-
(A) 0 (B) 2 (C) 4 (D) more than 4.

Paragraph for Questions 11 and 12

Consider a function $f(x) = \frac{x^3}{3} - \frac{x^2}{2} + x + \frac{1}{12}$.

On the basis of above information, answer the following questions :

11. The value of $\sum_{r=1}^{99} f\left(\frac{r}{100}\right)$ is -
(A) $\frac{95}{2}$ (B) $\frac{97}{2}$ (C) $\frac{99}{2}$ (D) $\frac{101}{2}$
12. The value of integral $\int_{1/7}^{6/7} f(f(x))dx$ is -
(A) $\frac{1}{14}$ (B) $\frac{3}{14}$ (C) $\frac{5}{14}$ (D) $\frac{7}{14}$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	○	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	●	●	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

-									
●	●	●	○	○	○	○	○	○	○
①	①	①	①	①	①	①	①	①	①
②	②	②	②	●	②	②	②	②	②
③	③	③	③	③	③	③	③	③	③
④	④	④	④	④	④	④	④	④	④
⑤	⑤	⑤	●	⑤	⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨	⑨

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A polynomial equation $x^4 + px^2 + qx + r = 0$; $p, q, r \in \mathbb{R} - \{0\}$; has 3 repeated real roots . If (p, r) lies on conic, then its semi latus rectum is
2. Let $\vec{a}, \vec{b}, \vec{c}$ are unit vectors and $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{c} \cdot \vec{a} = \cos \theta$; $\theta \in [0, \pi]$, then maximum value of θ is $k\pi$, where k is equal to
3. Given A and B are two non-singular matrices such that $B \neq I$, $A^5 = I$ and $AB^2 = BA$, then the least value of n for which $B^n = I$ is -
 (Where $n \in \mathbb{N}$)

Space for Rough Work

4. Let $A = \begin{bmatrix} \sqrt{3} & 1 & 0 \\ 1 & -\sqrt{3} & 0 \\ 0 & 0 & 2 \end{bmatrix}$ then the value of $\det(2A^T + AA^T + \text{adj}A)$ is equal to
5. Let f be a derivable function such that
 $f(2x + y^2) = f(2x) + f(y^2) + 10x^2y^2 + 5xy^4 - 4 \quad \forall x, y \in \mathbb{R}$, if $f(1) = 5$, then $f(2)$ is equal to
6. α, β, γ are roots of $x^3 + x^2 - 5x - 1 = 0$, then value of $[\alpha] + [\beta] + [\gamma]$ is
 (where $[.]$ denotes greatest integer function)

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has two sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I(i)	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
I(ii)	Paragraph Based (Single correct option)	6	+3 If only the bubble corresponding to the correct option is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18
II	Numerical Value Type (Up to second decimal place)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator